

RESEARCH EXPERIENCE

During my Ph.D., I concentrated on object detection within the context of autonomous driving, addressing the issues of out-of-distribution and domain shift.

- Instance-level Perception and Recognition** 2020-Present
 - Object Detection; Instance Segmentation
- Learning in the Open World with Out-of-distribution and Distribution Shift** 2020-2024
 - Domain Adaptation; Open-set Recognition; Open-Vocabulary Detection
- Learning with Latent Structure and Topology** 2020-2023
 - Causality; Graph-based Learning; Complex Networks
- Generative Models** 2023-Present
 - Video Generation; Image Manipulation; Discriminative Tasks

In addition, I have participated as a co-author in research on the following topics.

- 3D Vision** 2023-2024
 - Gaussian Splatting; 3D Point Cloud Segmentation; Stereo Vision
- Medical Imaging Analysis** 2021-Present
 - LLM Agent; MRI Brain Analysis; Tooth Analysis; Report Generation

EDUCATION

- The Chinese University of Hong Kong (CUHK)** Feb. 2024 - Dec. 2024
 - Postdoc, Electronic Engineering
 - Supervisor:** Prof. Yixuan Yuan
- City University of Hong Kong (CityU)** Aug. 2020 - Dec. 2023
 - Ph.D., Electrical Engineering
 - Supervisor:** Prof. Yixuan Yuan
 - Early Graduation
- Tianjin University (TJU)** Sep. 2016 - Jun. 2020
 - B.S., Electrical Engineering
 - GPA:** 3.83/4.0 (Top 2%)

UNIQUENESS

- Founder & Director of ScholaGO Education Technology Company Limited, Hong Kong [Link]**
- To gain a deeper understanding of technology, I founded a startup company to develop an innovative educational product. My ultimate goal is to develop valuable technology and products to improve the national happiness index.
- Research Highlights and Experience**
- My works were selected as **Oral Presentations** at AAAI-22, MICCAI-22, CVPR-22, and ICCV-23. Additionally, I was honored as a **Best Paper Finalist** at CVPR-22 and had my work featured as a **Cover Paper** in ACS Photonics. I embarked on my research journey in 2020 and successfully defended my PhD in 2023 without prior master's experience.

CONFERENCE PAPER

- CLIFF: Continual Latent Diffusion for Open-Vocabulary Object Detection**
Wuyang Li, Xinyu Liu, Jiayi Ma, Yixuan Yuan
European Conference on Computer Vision (**ECCV**), 2024
- Boosting Medical Image Analysis via Motion-Informed Generative Videos**
Wuyang Li, Xinyu Liu, Qiushi Yang, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2024
- Novel Scenes & Classes: Towards Adaptive Open-set Object Detection**
Wuyang Li, Xiaoqing Guo, Yixuan Yuan
IEEE International Conference on Computer Vision (**ICCV**), 2023, **Oral (1.8%)**
- Adjustment and Alignment for Unbiased Open Set Domain Adaptation**
Wuyang Li, Jie Liu, Bo Han, Yixuan Yuan
IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2023
- SIGMA: Semantic-complete Graph Matching for Domain Adaptive Object Detection**
Wuyang Li, Xinyu Liu, Yixuan Yuan
IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2022, **Best Paper Finalist (0.4%)**

6. **SCAN: Cross Domain Object Detection with Semantic Conditioned Adaptation**
Wuyang Li, Xinyu Liu, Xiwen, Yao, Yixuan Yuan
 The Association for the Advance of Artificial Intelligence (**AAAI**), 2022, **Oral (4.5%)**
7. **Joint polyp detection and segmentation with heterogeneous endoscopic data**
Wuyang Li, Chen Yang, Jie Liu, Xiaoqing Guo, Yixuan Yuan
 International Symposium on Biomedical Imaging (**ISBI**) Workshop, 2021
8. **When 3D Partial Points Meets SAM: Tooth Point Cloud Segmentation with Sparse Labels**
 Yifan Liu, Wuyang Li, Cheng Wang, Hui Chen, Yixuan Yuan
 Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2024, **Early Accept (14.0%)**
9. **DiffRect: Latent Diffusion Label Rectification for Semi-supervised Medical Image Segmentation**
 Xinyu Liu, Wuyang Li, Yixuan Yuan
 Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2024
10. **F2TNet: FMRI to T1w MRI Knowledge Transfer Network for Brain Multi-phenotype Prediction**
 Zhibin He, Wuyang Li, Yu Jiang, Zhihao Peng, Pengyu Wang, Xiang Li, Tianming Liu, Junwei Han, et al.
 Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2024
11. **LGS: A Light-weight 4D Gaussian Splatting for Efficient Surgical Scene Reconstruction**
 Hengyu Liu*, Yifan Liu*, Chenxin Li*, Wuyang Li, Yixuan Yuan
 Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2024
12. **Endora: Video Generation Models as Endoscopy Simulators**
 Chenxin Li, Hengyu Liu, Yifan Liu, Brandom Feng, Wuyang Li, Xinyu Liu, Zhen Chen, Jing Shao, Yixuan Yuan
 Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2024
13. **MRM: Masked Relation Modeling for Medical Image Pre-Training with Genetics**
 Qiushi Yang, Wuyang Li, Baopu, Li Yixuan Yuan
 IEEE International Conference on Computer Vision (**ICCV**), 2023
14. **H2GM: A Hierarchical Hypergraph Matching Framework for Brain Landmark Alignment**
 Zhibin He, Wuyang Li, Tuo Zhang, Yixuan Yuan
 Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2023
15. **Towards Robust Adaptive Object Detection under Noisy Annotations**
 Xinyu Liu, Wuyang Li, Qiushi Yang, Baopu Li, Yixuan Yuann
 IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2022
16. **Intervention & Interaction Federated Abnormality Detection with Noisy Clients**
 Xinyu Liu, Wuyang Li, Yixuan Yuan
 Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2022, **Oral**

JOURNAL PAPER

1. **Bridge the Gap Between Visual and Linguistic Comprehension for Zero-shot Semantic Segmentation**
 Xiaoqing Guo*, Wuyang Li*, Xinyu Liu, Yixuan Yuan (* Equal Contribution)
 International Journal of Computer Vision (**IJCV**), 2024, Under Revision
2. **SIGMA++: Improved Semantic-complete Graph Matching for Domain Adaptive Object Detection**
Wuyang Li, Xinyu Liu, Yixuan Yuan
 IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**), 2023
3. **SCAN++: Enhanced Semantic Conditioned Adaptation for Domain Adaptive Object Detection**
Wuyang Li, Xinyu Liu, Yixuan Yuan
 IEEE Transactions on Multimedia (**TMM**), 2022
4. **HTD: Heterogeneous Task Decoupling for Two-Stage Object Detection**
Wuyang Li, Zhen Chen, Baopu Li, Dingwen Zhang, Yixuan Yuan
 IEEE Transactions on Image Processing (**TIP**), 2021
5. **Stereo Vision Meta-Lens-Assisted Driving Vision**
 Xiaoyuan Liu, Wuyang Li, Takeshi Yamaguchi, Zihan Geng, Takuo Tanaka, Din Ping Tsai, Mu Ku Chen
 ACS Photonics, 2024, **Selected as Cover Paper**
6. **LLM-guided Decoupled Probabilistic Prompt for Continual Learning in Medical Image Diagnosis**
 Yiwen Luo, Wuyang Li, Chen Cheng, Xiang Li, Tianming Liu, Yixuan Yuan
 IEEE Transactions on Medical Imaging (**TMI**), 2024, Under Revision

7. **FM-APP: Foundation Model for Any Phenotype Prediction via fMRI to sMRI Knowledge Transfer**
Zhibin He, Wuyang Li, Yifan Liu, Xinyu Liu, Junwei Han, Tuo Zhang, Yixuan Yuan
IEEE Transactions on Medical Imaging (**TMI**), 2024, Under Review
8. **Medical Federated Learning with Joint Graph Purification for Noisy Label Learning**
Zhen Chen, Wuyang Li, Xiaohan Xing, Yixuan Yuan
Medical Image Analysis (**MIA**), 2023
9. **Decoupled Unbiased Teacher for Source-free Domain Adaptive Medical Object Detection**
Xinyu Liu, Wuyang Li, Yixuan Yuan
IEEE Transactions on Neural Networks and Learning Systems (**TNNLS**), 2023
10. **GRAB-Net: Graph-based Boundary-aware Network for Medical Point Cloud Segmentation**
Yifan Liu, Wuyang Li, Jie Liu, Hui Chen, Yixuan Yuan
IEEE Transactions on Medical Imaging (**TMI**), 2023

PREPRINT

1. **U-KAN Makes Strong Backbone for Medical Image Segmentation and Generation**
Chenxin Li*, Xinyu Liu*, Wuyang Li*, Cheng Wang*, Hengyu Liu, Yixuan Yuan (* Equal Contribution)
ArXiv preprint arXiv:2406.02918, 2024

RESEARCH EXPERIENCE

- **The Chinese University of Hong Kong** Feb. 2024 - Dec. 2024
Image/video generative models.
Supervisor: Prof. Yixuan Yuan
- **City University of Hong Kong** Aug. 2020 - Dec. 2023
Develop domain/noise-robust generic object detectors.
Supervisor: Prof. Yixuan Yuan
- **NUS (Suzhou) Research Institute (NUSRI)** Sep. 2019 - Jun. 2020
Develop a web-page-based Augmentation Reality (AR) framework.
Supervisor: Prof. Zhiying, Zhou

SELECTED AWARDS

- **Outstanding Academic Performance Award (OAPA), CityU** 2022-2023
Outstanding research achievements in the EE department.
- **Research Tuition Scholarship (RTS), CityU** 2022-2023
Outstanding research achievements in the EE department.
- **Ranked First in Polyp Detection Challenge** May 2021
3rd International Endoscopy Computer Vision Challenge and Workshop (EndoCV2021)
- **National Scholarship** 2017-2018
The Highest Honor for a Student in China (Top 2%)
- **Silver Award in Tianjin Advanced Mathematical Competition** Jun 2017
As a Selected Participant Representing Tianjin University (Top 5%)
- **First-class Student Awards** 2017-2019
Outstanding Students Selected in Tianjin University (Top 10%)

SERVICES

- **T-PAMI; IJCV; TIP; TNNLS; T-IV; T-ASE; PR** -
Journal Reviewer
- **NeurIPS-24; CVPR-23/24; ICCV-23; AAAI-23/24** -
Conference Reviewer